Lumina Royale Priority Reservation System

A fine dining restaurant like "Lumina Royale" has a waiting list based on customer priority. Every customer who has the highest priority, marked with a small priority number, will be served first. The problem should be solved using a double-linked list solution.

Input Format

- Line 1: N (number of operations)
- The next N lines each contain one of the following operations:
  - `ADD [Name] [Priority]`: Add a customer with the given name and priority number to the waiting list.
  - `SERVE`: Serve (remove) the customer with the highest priority (lowest priority number).
  - `NEXT`: Display the customer that will be served next (highest priority).
  - `COUNT`: Display the number of customers currently in the waiting list.
  - `DISPLAY`: Display all customers in the waiting list ordered by priority.
  - `DELETE [Name]`: Remove the customer with the given name from the waiting list.

Output Format

- `SERVE`: Print `Served customer: [Name] ([Priority])`, or `DOUBLE LINKED-LIST EMPTY.` if the list is empty.
- `NEXT`: Print `Next customer: [Name] ([Priority])`, or `DOUBLE LINKED-LIST EMPTY.` if the list is empty.
- `COUNT`: Print `Number of customers: [count]`, or `DOUBLE LINKED-LIST EMPTY.` if the list is empty.
- `DISPLAY`: Print `Waiting List:` followed by each customer as `[Name] ([Priority])` on separate lines, ordered by priority; or `DOUBLE LINKED-LIST EMPTY.` if the list is empty.
- `DELETE`: Print `Deleted customer: [Name] ([Priority])`, or `DOUBLE LINKED-LIST EMPTY.` if the list is empty.

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Constraints

- $1 \leq N \leq 50$
- Implement your algorithm in C using a double-linked list.
- The lower the priority value, the faster the customer is served.
- If two customers have the same priority, the customer who arrived first will be served first (FIFO).

Sample Input 1 (standard input)

```
3
ADD JOKO 2
ADD BUDI 1
DISPLAY
```

Sample Output 1 (standard output)

```
Waiting List:
BUDI (1)
JOKO (2)
```

Sample Input 2 (standard input)

```
12
ADD Alice 3
ADD Bob 1
ADD Clara 2
DISPLAY
NEXT
SERVE
NEXT
DELETE Clara
COUNT
DISPLAY
SERVE
DISPLAY
```

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Sample Output 2 (standard output)

```
Waiting List:
Bob (1)
Clara (2)
Alice (3)
Next customer: Bob (1)
Served customer: Bob (1)
Next customer: Clara (2)
Deleted customer: Clara (2)
Number of customers: 1
Waiting List:
Alice (3)
Served customer: Alice (3)
DOUBLE LINKED-LIST EMPTY.
```

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